

606025



Test Booklet No.

ENGLISH

Test Booklet Code

09

NEET RANKERS TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with blue/black ball point pen only.
- The test is of 3 hours 20 minutes duration and the Test Booklet contains 200 multiple-choice questions (four options with a single correct answer) from Physics, Chemistry and Biology (Botany and Zoology). 50 questions in each subject are divided into two Sections (A and B) as per details given below:
 - Section A shall consist of 35 (Thirty-five) Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
 - Section B shall consist of 15 (Fifteen) questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to attempt any 10 (Ten) questions out of 15 (Fifteen) in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- Use Blue/Black Ball Point Pen only for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The CODE for this Booklet is 09. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is NOT permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): _____

Roll Number : In figures _____

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Candidate's Signature: _____ Invigilator's Signature: _____

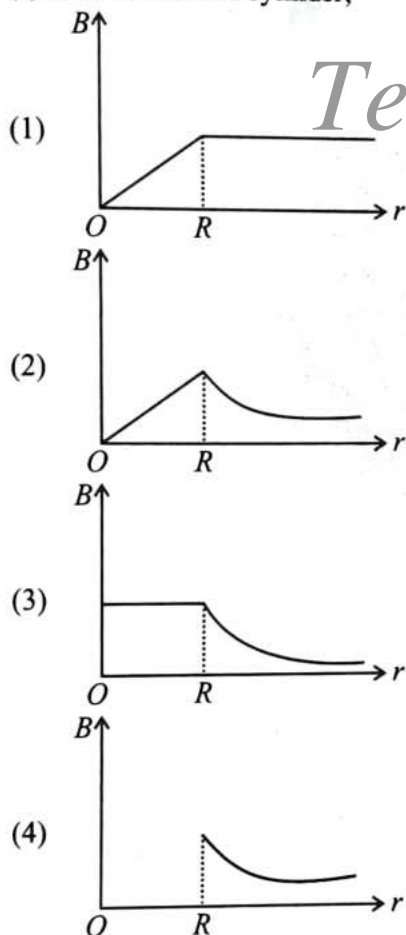
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Physics : Section-A (Q. No. 1 to 35)

1. Planck's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are three fundamental constants. Which of the following combinations of these has the dimension of length?

(1) $\sqrt{\frac{hc}{G}}$ (2) $\sqrt{\frac{Gc}{h^{3/2}}}$
 (3) $\sqrt{\frac{hG}{c^3}}$ (4) $\sqrt{\frac{hG}{c^{5/2}}}$

2. An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface. Choose the correct representation of magnetic field B as a function of radial distance (r) from the axis of cylinder;



3. The intervals measured by a clock given the following readings: 1.25 s, 1.24 s, 1.27 s, 1.21 s and 1.28 s. What is the percentage relative error in the observations?

(1) 4% (2) 16%
 (3) 1.6% (4) 2%

4. A ball is thrown vertically upward with an initial velocity of 150 m/s. The ratio of velocity after 3s and 5s is $\frac{x+1}{x}$. The value of x is {Take, $g = 10$

m/s²}
 (1) 10 (2) 5
 (3) 6 (4) -5

5. Given below are two statements;

Statement-I: When μ amount of an ideal gas undergoes adiabatic change from state (p_1, V_1, T_1) to state (p_2, V_2, T_2), then work done is $W = \frac{\mu R(T_2 - T_1)}{1 - \gamma}$, where $\gamma = \frac{C_p}{C_v}$ and $R =$

universal gas constant.

Statement-II: In the above case, when work is done on the gas, the temperature of the gas would rise.

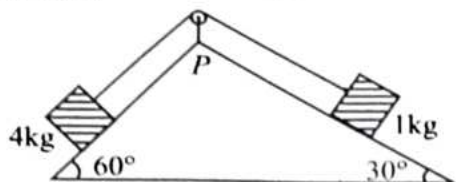
Choose the **correct** answer form the options given below.

- (1) Both statement-I and Statement-II are true.
 (2) Both Statement-I and Statement-II are false.
 (3) Statement-I is true but Statement-II are false.
 (4) Statement-I is false but Statement-II is true.

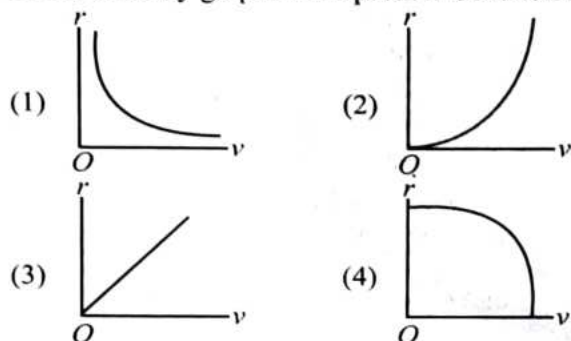
6. A projectile is given an initial velocity of $(\hat{i} + 2\hat{j})$ m/s, where $\hat{i}$ is along the ground and $\hat{j}$ is along the vertical. If $g = 10$ m/s², the equation of its trajectory is;

(1) $y = x - 5x^2$ (2) $y = 2x - 5x^2$
 (3) $4y = 2x - 5x^2$ (4) $4y = 2x - 25x^2$

7. As per given figure, a weightless pulley P is attached on double inclined frictionless surfaces. The tension in the string (massless) will be (if $g = 10 \text{ m/s}^2$)



- (1) $4(\sqrt{3} + 1) \text{ N}$
 (2) $(4\sqrt{3} + 1) \text{ N}$
 (3) $(4\sqrt{3} - 1) \text{ N}$
 (4) $4(\sqrt{3} - 1) \text{ N}$
8. A particle of mass m moves in a circular orbit under the central potential field, $U(r) = \frac{-C}{r}$, where C is a positive constant. The correct radius-velocity graph of the particle's motion is;



9. Given below are two statements, one is labelled as Assertion (A) and the other is labeled as Reason (R).

Assertion (A): If an electric dipole of dipole moment $30 \times 10^{-5} \text{ C-m}$ is enclosed by a closed surface, the net flux coming out of the surface will be zero.

Reason (R): Electric dipole consists of two equal and opposite charges.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

- (1) Both A and R are true and R is the correct explanation of A.
 (2) A is false but R is true.
 (3) Both A and R are true but R is not the correct explanation of A.
 (4) A is true but R is false.
10. Consider a system of two particles having masses m_1 and m_2 . If the particle of mass m_1 is pushed towards the centre of mass particles through a distance ' d ' by what distance would the particle of mass m_2 move so as to keep the centre of mass particles at the original position;

- (1) $\frac{m_1}{m_2} d$ (2) d
 (3) $\frac{m_1}{m_2}$ (4) $\frac{m_1}{m_1 + m_2} d$

11. A thin rod of length L and mass M is bent at its midpoint into two halves so that the angle between them is 90° . The moment of inertia of the bent rod about an axis passing through the bending point and perpendicular to the plane defined by the two halves of the rod is;

- (1) $\frac{\sqrt{2}ML^2}{24}$ (2) $\frac{ML^2}{24}$
 (3) $\frac{ML^2}{12}$ (4) $\frac{ML^2}{6}$

12. The kinetic energy of emitted electron is E when the light incident on the metal has wavelength λ . To double the kinetic energy, the incident light must have wavelength.

- (1) $\frac{hc}{E\lambda - hc}$ (2) $\frac{hc\lambda}{E\lambda + hc}$
 (3) $\frac{h\lambda}{E\lambda + hc}$ (4) $\frac{hc\lambda}{E\lambda - hc}$

13. In Li^{++} , electron in first Bohr orbit is excited to a level by a radiation of wavelength λ . When the ion gets de-excited to the ground state in all possible ways (including intermediate emissions), a total of six spectral lines are observed. What is the value of λ ? [Take, $h = 6.63 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ ms}^{-1}$]

- (1) 9.4 nm (2) 12.3 nm
(3) 10.8 nm (4) 11.4 nm

14. A circuit element X when connected to an AC supply of peak voltage 100 V gives a peak current of 5 A which is in phase with the voltage. A second element Y when connected to the same AC supply also gives the same value of peak current which lags behind the voltage by $\frac{\pi}{2}$. If X and Y are connected in series to the same supply then rms current in ampere?

- (1) $\frac{10}{\sqrt{2}}$ (2) $\frac{5}{\sqrt{2}}$
(3) $5\sqrt{2}$ (4) $\frac{5}{2}$

15. A uniform cylindrical rod of length L and radius r , is made from a material whose Young's modulus of elasticity equals Y . When this rod is heated by temperature T and simultaneously subjected to a net longitudinal compressional force F , its length remains unchanged. The coefficient of volume expansion, of the material of the rod, is (nearly) equal to;

- (1) $9F/(\pi r^2 Y T)$ (2) $6F/(\pi r^2 Y T)$
(3) $3F/(\pi r^2 Y T)$ (4) $F/(\pi r^2 Y T)$

16. A drop of liquid of density ρ is floating half immersed in a liquid of density σ and surface tension $7.5 \times 10^{-4} \text{ N cm}^{-1}$. The radius of drop in cm will be (Take, $g = 10 \text{ ms}^{-2}$)

- (1) $\frac{15}{\sqrt{2\rho - \sigma}}$ (2) $\frac{15}{\sqrt{\rho - \sigma}}$
(3) $\frac{3}{2\sqrt{\rho - \sigma}}$ (4) $\frac{3}{20\sqrt{2\rho - \sigma}}$

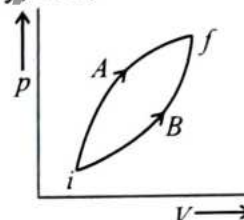
17. On a temperature scale X , the boiling point of water is $65^\circ X$ and the freezing point is $-15^\circ X$. Assume that the X scale is linear. The equivalent temperature corresponding to $-95^\circ X$ on the Fahrenheit scale would be;

- (1) -148°F (2) -112°F
(3) -48°F (4) -63°F

18. In a series L - C - R circuit, the inductive reactance (X_L) is 10Ω and the capacitive reactance (X_C) is 4Ω . The resistance (R) in the circuit is 6Ω . The power factor of the circuit is;

- (1) $\frac{1}{2}$ (2) $\frac{1}{2\sqrt{2}}$
(3) $\frac{1}{\sqrt{2}}$ (4) $\frac{\sqrt{3}}{2}$

19. Following figure shows two processes A and B for a gas. If ΔQ_A and ΔQ_B are the amount of heat absorbed by the system in two cases, and ΔU_A and ΔU_B are changes in internal energies respectively, then;



- (1) $\Delta Q_A > \Delta Q_B$, $\Delta U_A > \Delta U_B$
(2) $\Delta Q_A < \Delta Q_B$, $\Delta U_A < \Delta U_B$
(3) $\Delta Q_A > \Delta Q_B$, $\Delta U_A = \Delta U_B$
(4) $\Delta Q_A = \Delta Q_B$, $\Delta U_A > \Delta U_B$

20. A boy reaches the airport and finds that the escalator is not working. He walks up the stationary escalator in time t_1 . If he remains stationary on a moving escalator, then the escalator takes him up in time t_2 . The time taken by him to walk up on the moving escalator will be;

- (1) $\frac{t_1 t_2}{t_2 - t_1}$ (2) $\frac{t_1 + t_2}{2}$
(3) $\frac{t_1 t_2}{t_2 + t_1}$ (4) $t_2 - t_1$

21. A cylindrical tube open at both ends, has a fundamental frequency f , in air. The tube is dipped vertically in water so that half of it is in water. The fundamental frequency of the air-column is now;

(1) f (2) $f/2$
(3) $2f/4$ (4) $2f$

22. An ideal gas is expanding such that $pT^3 = \text{constant}$. The coefficient of volume expansion of the gas is;

(1) $1/T$ (2) $2/T$
(3) $4/T$ (4) $2/T$

23. Four charges equal to $-Q$ are placed at the four corners of a square and a charge q is at its centre. If the system is in equilibrium, the value of q is;

(1) $-\frac{Q}{4}(1+2\sqrt{2})$ (2) $\frac{Q}{4}(1+2\sqrt{2})$
(3) $-\frac{Q}{2}(1+2\sqrt{2})$ (4) $\frac{Q}{2}(1+2\sqrt{2})$

24. Two particles of equal mass(m) go around a circle of radius R under the action of their mutual gravitational attraction. Find the speed of the particle.

(1) $\frac{1}{2}\sqrt{\frac{Gm}{R}}$ (2) $\sqrt{\frac{4Gm}{R}}$
(3) $\frac{1}{2R}\sqrt{\frac{1}{Gm}}$ (4) $\sqrt{\frac{Gm}{R}}$

25. A string of length 1 m and mass 5 g is fixed at both ends. The tension in the string is 8.0 N. The string is set into vibration using an external vibrator of frequency 100 Hz. The separation between successive nodes on the string is close to;

(1) 16.6 cm (2) 33.3 cm
(3) 10.0 cm (4) 20.0 cm

26. A thermally insulated vessel contains an ideal gas of molecular mass M and ratio of specific heats is 1.4. Vessel is moving with speed v and is suddenly brought to rest. Assuming no heat is lost to the surrounding and vessel, temperature of the gas increases by (where, R = universal gas constant)

(1) $\frac{Mv^2}{7R}$ (2) $\frac{Mv^2}{5R}$
(3) $\frac{2Mv^2}{7R}$ (4) $\frac{7Mv^2}{5R}$

27. A $10 \mu\text{C}$ charge is divided into two parts and placed at 1 cm distance so that the repulsive force between them is maximum. The charges of the two parts are;

(1) $7\mu\text{C}, 3\mu\text{C}$ (2) $9\mu\text{C}, 1\mu\text{C}$
(3) $5\mu\text{C}, 5\mu\text{C}$ (4) $8\mu\text{C}, 2\mu\text{C}$

28. The time period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle, which moves without friction down an inclined plane of inclination α , is given by;

(1) $2\pi\sqrt{L/(g \cos \alpha)}$ (2) $2\pi\sqrt{L/(g \sin \alpha)}$
(3) $2\pi\sqrt{L/g}$ (4) $2\pi\sqrt{L/(g \tan \alpha)}$

29. Two thin wire rings each having a radius R are placed at a distance d apart with their axes coinciding. The charges on the two rings are $+q$ and $-q$. The potential difference between the centres of the two rings is;

(1) $\frac{qR}{4\pi\epsilon_0 d^2}$
(2) $\frac{q}{2\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$
(3) zero
(4) $\frac{qR}{4\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}} \right]$

30. A nucleus disintegrates into two nuclear parts which have their velocities in the ratio 2 : 1. The ratio of their nuclear sizes will be;

(1) $2^{1/3} : 1$ (2) $1 : 2^{1/3}$
(3) $3^{1/3} : 1$ (4) $1 : 2^{1/3}$

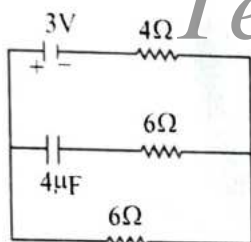
31. A wire of resistance R_1 is drawn out so that its length becomes twice of its original length. The ratio of new resistance to original resistance is;

(1) 9 : 1 (2) 1 : 9
(3) 4 : 1 (4) 3 : 1

32. A copper rod of cross-sectional area A carries a uniform current I through it. At temperature T , if the volume charge density of the rod is ρ , how long will the charges take to travel a distance d ?

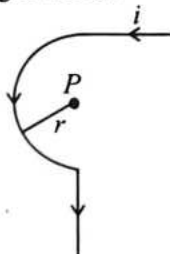
(1) $\frac{\rho A d}{I}$ (2) $\frac{\rho d Q}{IT}$
(3) $\frac{2\rho d Q}{I}$ (4) $\frac{2\rho d Q}{IT}$

33. In the network shown below, the charge accumulated in the capacitor in steady state will be;



(1) 12 μC (2) 4.8 μC
(3) 7.2 μC (4) 103 μC

34. Find the magnetic field at the point P in figure. The curved portion is a semicircle connected to two long straight wires.



(1) $\frac{\mu_0 i}{2r} \left(\frac{1}{2} + \frac{1}{2\pi} \right)$ (2) $\frac{\mu_0 i}{2r} \left(1 + \frac{1}{\pi} \right)$
(3) $\frac{\mu_0 i}{2r} \left(\frac{1}{2} + \frac{1}{\pi} \right)$ (4) $\frac{\mu_0 i}{2r} \left(1 + \frac{1}{\pi} \right)$

35. The magnetic field at the centre of a circular coil of radius r , due to current I flowing through it, is B . The magnetic field at a point along the axis at a distance $r/2$ from the centre is;

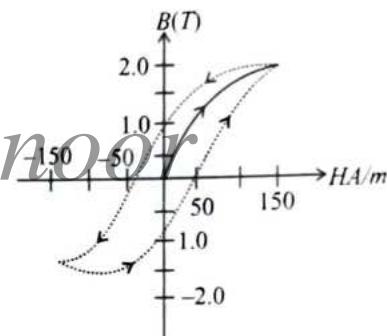
(1) $B/2$ (2) $2B$
(3) $\left(\frac{2}{\sqrt{5}} \right)^3 B$ (4) $\left(\frac{2}{\sqrt{3}} \right)^3 B$

Physics : Section-B (Q. No. 36 to 50)

36. An equation is given here $\left(P + \frac{a}{V^2} \right) = b \frac{\theta}{V}$ where P = Pressure, V = Volume and θ = Absolute temperature. If a and b are constants, the dimensions of a will be;

(1) $[\text{ML}^{-5} \text{T}^{-1}]$ (2) $[\text{ML}^5 \text{T}^1]$
(3) $[\text{ML}^5 \text{T}^{-2}]$ (4) $[\text{ML}^{-1} \text{T}^2]$

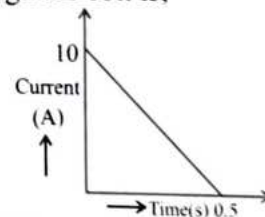
- 37.



The figure gives experimentally measured B versus H variation in a ferromagnetic material. The retentivity, coercivity and saturation respectively of the material are;

(1) 1.0 T, 50 A/m and 1.5 T
(2) 150 T, 1.0 A/m and 1.5 T
(3) 1.5 T, 50 A/m and 1.5 T
(4) 1.5 T, 50 A/m and 1.5 T

38. In a coil of resistance 100 Ω , a current is induced by changing the magnetic flux through it as shown in the figure. The magnitude of change in flux through the coil is;

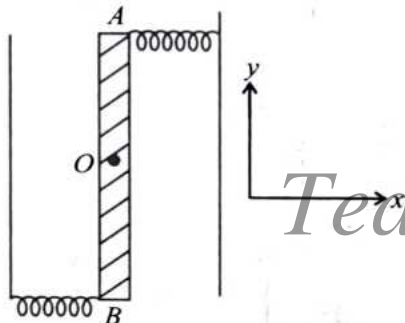


(1) 225 Wb (2) 250 Wb
(3) 275 Wb (4) 200 Wb

39. A capacitor of capacitance $150.0 \mu\text{F}$ is connected to an alternating source of emf given by;
 $E = 36 \sin(120\pi t) \text{ V}$. The maximum value of current in the circuit is approximately equal to;

- (1) 2 A (2) $\sqrt{2} \text{ A}$
 (3) $2\sqrt{2} \text{ A}$ (4) $1/\sqrt{2} \text{ A}$

40. Two light identical springs of spring constant k are attached horizontally at the two ends of a uniform horizontal rod AB of length l and mass m . The rod is pivoted at its centre ' O ' and can rotate freely in horizontal plane. The other ends of the two springs are fixed to rigid supports as shown in figure.



The rod is gently pushed through a small angle and releases. The frequency of resulting oscillation is;

- (1) $\frac{1}{2\pi} \sqrt{\frac{2k}{m}}$ (2) $\frac{1}{2\pi} \sqrt{\frac{3k}{m}}$
 (3) $\frac{1}{2\pi} \sqrt{\frac{6k}{m}}$ (4) $\frac{1}{2\pi} \sqrt{\frac{k}{m}}$

41. Two materials having coefficients of thermal conductivity ' $3K$ ' and ' K ' and thickness ' d ' and ' $3d$ ' respectively, are joined to form a slab. The temperatures of the outer surfaces are ' θ_2 ' and ' θ_1 ' respectively, ($\theta_2 > \theta_1$). The temperature at the interface is;

- (1) $\frac{\theta_2 + \theta_1}{2}$ (2) $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$
 (3) $\frac{\theta_1}{6} + \frac{5\theta_2}{6}$ (4) $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$

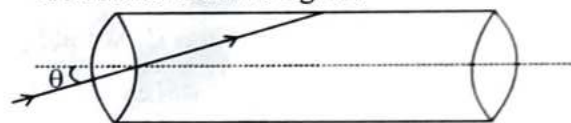
42. Match List-I with List-II.

List-I

List-II

- | | |
|----------------------------------|---|
| A. Gauss's law in electrostatics | I. $\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\phi_B}{dt}$ |
| B. Faraday's law | II. $\oint \mathbf{B} \cdot d\mathbf{A} = 0$ |
| C. Gauss's law in magnetism | III. $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 i_0 + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$ |
| D. Ampere-Maxwell law | IV. $\oint \mathbf{E} \cdot d\mathbf{s} = \frac{q}{\epsilon_0}$ |
- (1) A-I, B-III, C-IV, D-II
 (2) A-IV, B-I, C-II, D-III
 (3) A-I, B-IV, C-II, D-III
 (4) A-III, B-II, C-I, D-IV

43. A transparent solid cylinder rod has a refractive index of $2/\sqrt{3}$. It is surrounded by air. A light ray is incident at the mid-point of one end of the rod as shown in the figure.



The incident angle θ for which the light ray grazes along the wall of the rod is;

- (1) $\sin^{-1}\left(\frac{1}{2}\right)$ (2) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$
 (3) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (4) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

44. Two coherent light sources having intensity in the ratio $2x$ produce an interference pattern. The

ratio $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$.

- (1) $\frac{2\sqrt{2x}}{x+1}$ (2) $\frac{\sqrt{2x}}{2x+1}$
 (3) $\frac{\sqrt{2x}}{x+1}$ (4) $\frac{2\sqrt{2x}}{2x+1}$

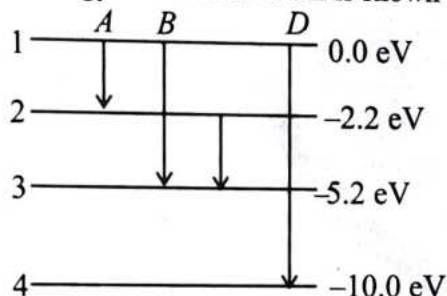
45. A thin circular ring of mass M and radius r is rotating about its axis with constant angular velocity ω . The objects each of mass m are attached gently to the opposite ends of a diameter of the ring. The ring now rotates with angular velocity given by;

(1) $\frac{(M+2m)\omega}{2m}$ (2) $\frac{2m\omega}{M+2m}$
 (3) $\frac{(M+2m)\omega}{M}$ (4) $\frac{M\omega}{M+2m}$

46. Electron beam used in an electron microscope, when accelerated by a voltage of 20 kV, has a de-Broglie wavelength of λ_0 . If the voltage is increased to 40 kV, then the de-Broglie wavelength associated with the electron beam would be;

(1) $9\lambda_0$ (2) $\frac{\lambda_0}{\sqrt{2}}$
 (3) $\frac{\lambda_0}{2}$ (4) $3\lambda_0$

47. The energy levels of an atom is shown in figure.



Which one of these transitions will result in the emission of a photon of wavelength 124.1 nm?

(Given, $h = 6.62 \times 10^{-34}$ J-s)

- (1) A (2) C
 (3) D (4) B

48. Kepler's third law states that square of period of revolution (T) of a planet around the sun, is proportional to third power of average distance r between sun and planet, i.e., $T^2 = Kr^3$ here K is constant. If the masses of sun and planet are M and m respectively then as per Newton's law of

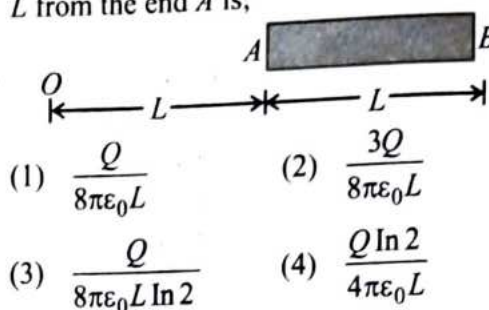
gravitation force of attraction between them is

$F = \frac{GMm}{r^2}$ here G is gravitational constant. The

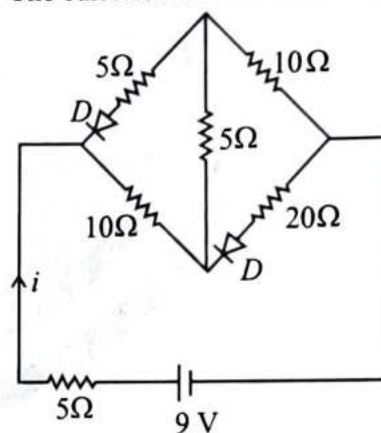
relation between G and K is described as;

- (1) $GMK = 4\pi^2$ (2) $K = G$
 (3) $K = \frac{1}{G}$ (4) $GM = 4\pi^2$

49. A charge Q is uniformly distributed over a long rod AB of length L as shown in the figure. The electric potential at the point O lying at distance L from the end A is;



50. The current i in the network is;



- (1) 0.2 A (2) 0 A
 (3) 0.6 A (4) 0.3 A

Chemistry : Section-A (Q. No. 51 to 85)

51. The osmotic pressure of a solution is given by; (whereas π = Osmotic pressure C = concentration)

(1) $\pi = \frac{RT}{C}$ (2) $\pi = \frac{CT}{R}$
 (3) $\pi = \frac{RC}{T}$ (4) $\frac{\pi}{C} = RT$

52. The oxidation number of middle carbon in the carbon (C_3O_2) sub-oxide is:
 (1) -2 (2) +4
 (3) $+\frac{4}{3}$ (4) Zero
53. The reaction between phenol and chloroform in presence of aqueous NaOH is:
 (1) Nucleophilic substitution reaction.
 (2) Electrophilic addition reaction.
 (3) Electrophilic substitution reaction.
 (4) Nucleophilic addition reaction.
54. Correct statements are:
 A. Fibrous proteins are generally insoluble in water
 B. Insulin and albumins are examples of fibrous proteins.
 C. Globular proteins are usually soluble in water.
 D. Keratin and myosin are the examples of globular proteins.
 (1) A and C (2) B, C and D
 (3) A and D (4) B and C
55. Incorrect statement(s) is/are
 A. Sulphur has less negative electron gain enthalpy than oxygen atom..
 B. Metallic character increases from oxygen to polonium.
 C. The thermal stability of hydrides increases from H_2O to H_2Po .
 D. The acidic character increases from H_2O to H_2Te .
 (1) A and B (2) B, C and D
 (3) A and C (4) Only B
56. Among the following species which is square pyramidal?
 (1) I_3^- (2) BrF_5
 (3) H_2O (4) CH_4
57. Which of the following pairs has the same size?
 (1) Fe^{2+} , Ni^{2+} (2) Zr^{4+} , Ti^{4+}
 (3) Zr^{4+} , Hf^{4+} (4) Zn^{2+} , Hf^{4+}

58. Match List-I with List-II.

List-I (Thermodynamic properties)	List-II (Condition of process)
(A) $\Delta U = q_v$	(I) Entropy change at equilibrium
(B) $\Delta H = q_p$	(II) Heat change at constant volume
(C) $\Delta S = 0$	(III) Process is non spontaneous
(D) $\Delta G > 0$	(IV) Heat change at constant pressure

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
 (2) A-IV, B-III, C-II, D-I
 (3) A-II, B-III, C-IV, D-I
 (4) A-I, B-IV, C-III, D-II

59. For the reaction $A + B \rightleftharpoons C + D$, the E_a for the forward reaction is 32 KJ mol^{-1} and E_a for the reverse reaction is 58 KJ mol^{-1} . Heat of reaction, ΔH is:
 (1) 52 KJ mol^{-1} (2) -52 KJ mol^{-1}
 (3) 26 KJ mol^{-1} (4) -26 KJ mol^{-1}

60. IUPAC name of compound
 $CH_3 - O - CH - CH_3$ is;
 $\quad \quad \quad |$
 $\quad \quad \quad CH_3$
 (1) 2-ethoxypropane
 (2) 2-methoxy-2-methylethane
 (3) 1-methoxyethane
 (4) 2-methoxypropane

61. Which of the following ions are colourless.

- A. Sc^{3+} B. Zn^{2+}
 C. Ti^{4+} D. Cu^{2+}

Choose **most appropriate** answer from the given options below:

- (1) A and B only
 (2) A, C and D only
 (3) A, B and C only
 (4) A, B, C and D

62. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: First ionisation enthalpy of nitrogen is more than oxygen.

Reason R: Nitrogen exactly have half filled electronic configuration.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is NOT the correct explanation of A.
63. Which alcohol produces turbidity with Lucas reagent most slowly?
- (1) Butan-2-ol
 - (2) 2-methylpropan-2-ol
 - (3) Isobutyl alcohol
 - (4) propan-2-ol

64. Given below are two statements:
Statement I: CN^- is ambidentate ligand.
Statement II: CN^- is bidentate ligand.
 In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

65. Given below are two statements:
Statement I: Boiling point of H_2O is more than H_2S .

Statement II: H_2S is more acidic than H_2O .

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

66. For which subshell value of $(n + l)$ is 5?

- (1) 4s
- (2) 5p
- (3) 3d
- (4) 4f

67. Match List-I with List-II.

List-I (Reaction)	List-II (Catalyst/Reagent)
(A) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH} \rightarrow \text{CH}_3-\text{CH}=\text{CH}_2$	(I) KMnO_4 in acidic medium
(B) $\text{CH}_2=\text{CH}_2 \rightarrow \text{CH}_3-\text{CH}_3$	(II) H_2SO_4 (conc) at 443K
(C) $\text{CH}_3-\text{CH}_2-\text{OH} \rightarrow \text{CH}_3-\text{CHO}$	(III) H_2/Ni
(D) $\text{CH}_3-\text{CH}_2-\text{OH} \rightarrow \text{CH}_3\text{COOH}$	(IV) Cu at 573K

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-III, C-IV, D-II
- (3) A-IV, B-II, C-III, D-I
- (4) A-II, B-III, C-IV, D-I

68. How will you convert butan-2-one to propanoic acid?

- (1) Tollen's reagent
- (2) Fehling solution
- (3) $\text{NaOH}/\text{I}_2/\Delta$ and H^+
- (4) $\text{NaOH}/\text{NaI}/\text{H}^+$

69. If I_2 and NaOH is heated with organic compound:

- A. Propanone gives iodoform reaction.
- B. Formic acid gives iodoform reaction.
- C. Propanol gives iodoform reaction.

Choose the most appropriate answer from the options given below:

- (1) A only
- (2) A and B only
- (3) A and C only
- (4) A, B and C

70. Match List-I with List-II.

List-I (Compound)	List-II (Number of σ and π bond between carbon atoms)
(A) Cyclopropene	(I) Six σ bond and one π bond
(B) Cyclobutane	(II) Three σ bond and one π bond
(C) Cyclohexene	(III) Four σ bond
(D) Cyclobutadiene	(IV) Four σ bond and two π bond

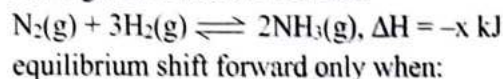
Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
 (2) A-II, B-III, C-I, D-IV
 (3) A-III, B-IV, C-II, D-I
 (4) A-I, B-III, C-IV, D-II

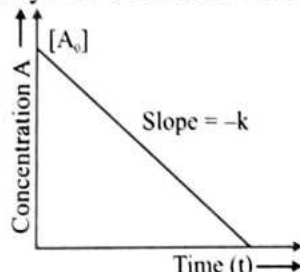
71. Thermal stability of hydrides of 15th group follows the order:

- (1) $\text{SbH}_3 < \text{AsH}_3 < \text{PH}_3 < \text{NH}_3$
 (2) $\text{NH}_3 < \text{SbH}_3 < \text{AsH}_3 < \text{PH}_3$
 (3) $\text{NH}_3 < \text{AsH}_3 < \text{PH}_3 < \text{SbH}_3$
 (4) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$

72. For a given reversible reaction



- (1) temperature increases.
 (2) pressure increases.
 (3) pressure decreases.
 (4) catalyst is added.

73. Consider the reaction $\text{A} \rightarrow \text{P}$. The change in concentration of A with time has been shown graphically. The order of the reaction is:

- (1) zero order (2) fractional order
 (3) first order (4) second order

74. A reaction is $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$. Initially we start with equal concentration of A and B. At equilibrium the moles of C is two times of A. What is the equilibrium constant of the reaction?

- (1) 4 (2) 2
 (3) $\frac{1}{4}$ (4) $\frac{1}{2}$

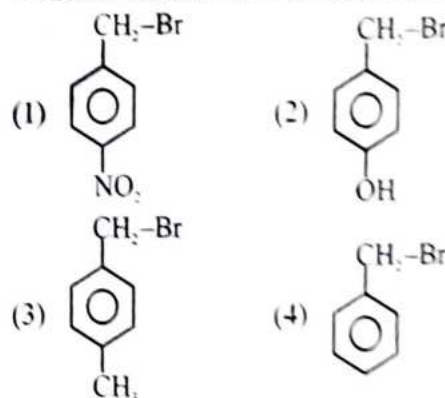
75. Given below are two statements:

Statement I: Melting point of n-pentane is less than neopentane.

Statement II: With branching, neopentane becomes symmetrical therefore packing efficiency increase.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

76. Which is most reactive towards $\text{S}_\text{N}1$ reaction?77. Calculate the radius ratio of 2nd excited state of H and 1st excited state of Li^{2+}

- (1) 9 : 4 (2) 27 : 4
 (3) 16 : 9 (4) 25 : 9

78. Which of the following gives fehling solution test?

- (1) $\text{CH}_3\text{---CH}_2\text{---OH}$ (2) CH_3COCH_3
 (3)  (4) $\text{CH}_3\text{---CHO}$

79. Impure sample of naphthalene can be purified by:

- (1) Distillation
 (2) Chromatography
 (3) Crystallisation
 (4) Sublimation

80. Match List-I with List-II.

List-I Coordination complex	List-II Number of unpaired electrons
(A) $[\text{Cr}(\text{CN})_6]^{3-}$	(I) 5
(B) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(II) 3
(C) $[\text{FeCl}_6]^{3-}$	(III) 2
(D) $[\text{Ni}(\text{NH}_3)_6]^{2+}$	(IV) 4

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
 (2) A-IV, B-III, C-II, D-I
 (3) A-II, B-IV, C-I, D-III
 (4) A-II, B-I, C-IV, D-III

81. Given below are two statements:

Statement I: Lower aliphatic amines are water soluble.

Statement II: Solubility of amines decreases with increase in molar mass.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

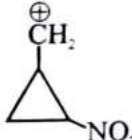
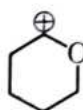
82. In which of the following the energy change corresponds to first ionisation potential only:

- (1) $\text{X(g)} \rightarrow \text{X}^+(\text{g}) + \text{e}^-$
 (2) $\text{X}_2(\text{g}) \rightarrow \text{X}^+(\text{g}) + \text{e}^-$
 (3) $\text{X(s)} \rightarrow \text{X}^+(\text{g}) + \text{e}^-$
 (4) $\text{X(aq)} \rightarrow \text{X}^+(\text{aq}) + \text{e}^-$

83. Which of the following contains the least number of molecules?

- (1) 4.4 g CO_2 (2) 3.4 g NH_3
 (3) 1.6 g CH_4 (4) 3.2 g SO_2

84. Which of the following carbocation is least stable.

- (1)  (2) 
 (3)  (4) 

85. 171 g of cane sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is dissolved in 1 litre of water. The molarity of the solution is:

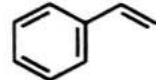
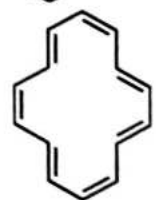
- (1) 2.0 M (2) 1.0 M
 (3) 0.5 M (4) 0.25 M

Chemistry : Section-B (Q. No. 86 to 100)

86. An oxide of metal M has 40% by mass of oxygen. Metal M has atomic mass of 24. The empirical formula of the oxide is: (atomic mass of oxygen = 16)

- (1) M_2O (2) M_2O_3
 (3) MO (4) M_3O_4

87. Which of the following are aromatic?

- (A)  (B) 
 (C)  (D) 

- (1) B and D only (2) A and C only
 (3) A and B only (4) C and D only



88. Match List-I with List-II.

List-I (Cell/Battery)	List-II (Overall reaction)
(A) Dry cell	(I) $\text{Zn(Hg)} + \text{HgO(s)} \rightarrow \text{ZnO(s)} + \text{Hg(l)}$
(B) Mercury cell	(II) $\text{Zn} + 2\text{NH}_4\text{Cl} + \text{MnO}_2 \rightarrow \text{ZnCl}_2 + 2\text{MnO(OH)} + 2\text{NH}_3$
(C) Lead storage Battery	(III) $\text{Cd(s)} + \text{Ni(OH)}_2\text{(s)} \rightarrow \text{CdO(s)} + 2\text{Ni(OH)}_2 + \text{H}_2\text{O(l)}$
(D) Nickel-cadmium cell	(IV) $\text{Pb(s)} + \text{PbO}_2\text{(s)} + 2\text{H}_2\text{SO}_4\text{(aq)} \rightarrow 2\text{PbSO}_4\text{(s)} + 2\text{H}_2\text{O(l)}$

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
 (2) A-IV, B-III, C-II, D-I
 (3) A-II, B-I, C-IV, D-III
 (4) A-II, B-I, C-III, D-IV

89. The product formed when But-2-ene treated with potassium permagnate in acidic medium at high temperature is:

- (1) Acetone + CO_2
 (2) Acetic acid
 (3) Acetic acid + CO_2
 (4) Acetic acid + acetone

90. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Acetaldehyde is more reactive than Ketone in nucleophilic addition reaction.

Reason R: Due to +I effect of two alkyl groups in ketone, electron density across carbonyl carbon increase therefore reactivity decreases. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.

91. In the reaction,
 $2\text{KMnO}_4 + 10\text{FeSO}_4 + 8\text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{SO}_4 + 2\text{MnSO}_4 + 5\text{Fe}_2(\text{SO}_4)_3 + 8\text{H}_2\text{O}$, what will be the equivalent weight of FeSO_4 ?

(Fe = 56u, Mn = 55u, S = 16u, K = 39u)

- (1) 152 (2) 75.5
 (3) 302 (4) 113.25

92. **Correct** order of Bond angle is:

- (1) $\text{SO}_3 > \text{SO}_2 > \text{H}_2\text{O}$
 (2) $\text{SO}_3 > \text{H}_2\text{O} > \text{SO}_2$
 (3) $\text{SO}_2 > \text{SO}_3 > \text{H}_2\text{O}$
 (4) $\text{H}_2\text{O} > \text{SO}_3 > \text{SO}_2$

93. The group reagent of first group basic radicals is:

- (1) Dil. H_2SO_4 (2) Dil. HCl
 (3) Conc. H_2SO_4 (4) Conc. HCl

94. Which is Finkelstein reaction?

- (1) $\text{R-X} + \text{NaI} \xrightarrow{\text{Acetone}} \text{R-I} + \text{NaX} \text{ (X = Cl, Br)}$
 (2) $\text{R-X} + \text{AgF} \longrightarrow \text{R-F} + \text{NaX} \text{ (X = Cl, Br)}$
 (3) $\text{Ar-NH}_2 \xrightarrow[273-278\text{K}]{\text{NaNO}_2 + \text{HX}} \text{Ar-N}_2^+ \text{X}^-$
 (4) $\text{R-OH} + \text{HCl} \xrightarrow{\text{ZnCl}_2} \text{R-Cl} + \text{H}_2\text{O}$

95. The rates of a certain reaction $\left(\frac{dc}{dt}\right)$ at different times are as follows:

Time	Rate (mole litre ⁻¹ s ⁻¹)
0	2.8×10^{-2}
10	2.78×10^{-2}
20	2.81×10^{-2}
30	2.79×10^{-2}

The reaction is:

- (1) Zero order (2) First order
 (3) Second order (4) Third order

96. In the beginning of the reaction, $\text{A} \rightleftharpoons \text{B}$, 4 moles of A are taken, out of which 0.8 moles gets dissociated. What is the degree of dissociation of A?

- (1) 1 (2) 0.125
 (3) 0.2 (4) 3

97. When 229 J of heat is supplied at constant pressure to 3 mol Ar(g), the temperature of the sample is increased by 2.55K. Calculate the molar heat capacity at constant volume:
 (1) $30 \text{ kJ K}^{-1} \text{ mol}^{-1}$ (2) $30 \text{ J K}^{-1} \text{ mol}^{-1}$
 (3) $21.7 \text{ J K}^{-1} \text{ mol}^{-1}$ (4) $21.7 \text{ kJ K}^{-1} \text{ mol}^{-1}$
98. 5 amperes current is passed through a solution of zinc sulphate for 40 minutes. The amount of zinc deposited at the cathode is;
 (1) 40.65 g (2) 4.065 g
 (3) 0.4065 g (4) 65.04 g
99. The **correct** reactivity order of alcohols towards HBr will be:
- I. $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_3 \\ | \\ \text{OH} \end{array}$
 II. $\text{CH}_3 - \text{CH}_2 - \text{OH}$
 III. $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_3 \\ | \\ \text{OH} \end{array}$
- (1) $\text{II} > \text{III} > \text{I}$ (2) $\text{I} > \text{III} > \text{II}$
 (3) $\text{III} > \text{II} > \text{I}$ (4) $\text{II} > \text{I} > \text{III}$
100. The elements present in 4f is also known as;
 (1) lanthanides. (2) rare earths elements
 (3) lanthanones. (4) All the above

Botany : Section-A (Q. No. 101 to 135)

101. Choose the **incorrect** statement for the universal rules of binomial nomenclature.
 (1) Biological names are generally in Latin or latinised.
 (2) The first word in a biological name represents the specific epithet while the second component denotes the genus.
 (3) Both the words in a biological name, when handwritten, are separately underlined, or printed in italics to indicate their Latin origin.
 (4) It was proposed by Carolus Linnaeus.

102. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: M phase is also known as equational division.
Reason R: M phase produces daughter cells with same number of chromosomes as parent cell.
 In the light of the above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
103. Mark the following statements as true (T) or false (F) and select the correct option.
 I. In mustard, filaments of stamens are of similar lengths.
 II. In Cruciferae, the outer two stamens are short, and the inner four stamens are long.
 III. The fruit of the china rose is a capsule.
 IV. Marigold, sunflower, and *Chrysanthemum* are members of the family Malvaceae.
 (1) I – False, II – True, III – True, IV – False
 (2) I – True, II – False, III – True, IV – False
 (3) I – False, II – False, III – False, IV – True
 (4) I – True, II – True, III – False, IV – True

104. Mesosome which is formed by the extensions of plasma membranes into the cell does **not** help in:
 (1) formation of cell wall.
 (2) replication of DNA.
 (3) respiration.
 (4) storing pigments.
105. In animal cells, lipid-like steroidal hormones are synthesised in:
 (1) rough endoplasmic reticulum.
 (2) golgi bodies.
 (3) lysosomes.
 (4) smooth endoplasmic reticulum.

106. _____ is the type of pollination which is a cross pollination functionally but self-pollination genetically.

- (1) Autogamy (2) Geitonogamy
(3) Xenogamy (4) Cleistogamy

107. Identify the **mismatched** pair w.r.t the steps of EMP pathway.

(1)	3-phosphoglyceraldehyde is converted to 1, 3-bisphosphoglycerate	NADH + H ⁺ is formed
(2)	Conversion of glucose into glucose 6-phosphate	ADP is utilised
(3)	Fructose 6-phosphate to fructose 1, 6-bisphosphate	ATP is utilised
(4)	Conversion of BPGA to 3-phosphoglyceric acid	ATP is formed

108. Identify the plant?



Terrestrial habitat

- (1) Larkspur
(3) Coriander



Water habitat

- (2) Buttercup
(4) Cotton

109. Three water samples were collected; drinking water, untreated sewage water and secondary effluent. The samples were labelled A, B and C but laboratory attendant did not note the source of sample. The BOD values of the samples recorded is 20 mg/L, 2 mg/L and 400 mg/L respectively. Using this data, select the correct source of water for the labels A, B and C

- (1) A - Secondary effluent
B - Drinking water
C - Untreated sewage
(2) A - Untreated sewage
B - Drinking water
C - Secondary effluent
(3) A - Drinking water
B - Secondary effluent
C - Untreated sewage
(4) A - Untreated sewage
B - Secondary effluent
C - Drinking water

110. Identify the cell which is **not** devoid of protoplasm.

- (1) Vessels
(2) Tracheids
(3) Xylem Parenchyma
(4) Sclerenchyma

111. Read the following statements ((a) - (e)) and answer the question which follows them

- (a) Dragonflies are used for controlling mosquitoes.
(b) Rumen of cattle is rich in methanogens.
(c) Increase in organic pollution increase BOD.
(d) Man-made technology for treatment of sewage is better than microbial treatment.
(e) Lipases are commercially used in detergents.

How many of the above statements are correct?

- (1) One (2) Two
(3) Three (4) Four

112. Identify the **incorrect** statement w.r.t factors affecting photosynthesis.

- (1) The internal factors which affect photosynthesis are dependent on the genetic predisposition of the plant.
(2) Despite the presence of a green leaf and optimal light and CO₂ conditions, the plant may not photosynthesis if the temperature is very low.
(3) Increase in concentration of CO₂ upto 0.05 per cent can cause an increase in CO₂ fixation rates.
(4) At low light conditions both C₃ and C₄ plants respond to high CO₂ conditions.

113. In the process of meiosis-I, terminalisation of chiasmata occurs during _____ stage.

- (1) leptotene (2) diakinesis
(3) zygotene (4) diplotene

114. Identify the classes of fungi in which mycelium is branched and septate & choose the **correct** option.

- (1) Phycomycetes and ascomycetes
(2) Basidiomycetes and phycomycetes
(3) Ascomycetes and deuteromycetes
(4) Deuteromycetes and phycomycetes

115. Given below are two statements:
Statement I: The term linkage is used to describe the physical association of genes on a chromosome.
Statement II: The term recombination is used to describe the generation of non-parental gene combinations.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
- Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.
 - Both Statement I and Statement II are correct.
 - Both Statement I and Statement II are incorrect.
116. If glycolysis starts with sucrose as the substrate then the number of molecule(s) of pyruvic acid formed is/are:
- 1
 - 2
 - 3
 - 4
117. Given below are two statements:
Statement I: In some organisms, karyokinesis is not followed by cytokinesis as a result of which a multinucleate condition arises leading to the formation of syncytium.
Statement II: Cell growth restores the ratio between the nucleus and the cytoplasm.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
- Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.
 - Both Statement I and Statement II are correct.
 - Both Statement I and Statement II are incorrect.
118. Read the given statements and choose the **correct** option w.r.t the group of plants.
- Grow in moist shaded areas.
 - Can live in soil but are dependent on water for sexual reproduction.

- The plant body is thallus-like and prostrate or erect.
 - The main plant body is gametophyte.
 - They prevent soil erosion.
- Algae
 - Pteridophytes
 - Bryophytes
 - Gymnosperms

119. The male gametes of rice plant have 10 chromosomes in their nucleus. The chromosome number in the female gamete, zygote and the cells of the seedling will be, respectively:
- 10, 20 and 10.
 - 20, 10 and 10.
 - 10, 20 and 20.
 - 20, 10 and 20.

120. Identify the **incorrect** statements.
- Dominance is an autonomous feature of a gene.
 - In an individual only two alleles can be present, multiple alleles can be found only when population studies are made.
 - Codominance is shown by human blood groups.
 - Starch synthesis in pea seed is controlled by single gene but it produces seeds in three sizes large, intermediate & small due to presence of multiple alleles.

Choose the *most appropriate* answer from the options given below:

- A and B
- C and D
- A and D
- A and C

121. Match List-I with List-II.

List-I	List-II
(A) Hershey and Chase	(I) Proved that DNA is the genetic material
(B) Watson and Crick	(II) Proposed double helix model for the structure of DNA
(C) Avery, MacLeod and McCarty	(III) Discovered that proteases and RNases did not affect transformation
(D) Griffith	(IV) Transformation in bacteria

Choose the **correct** answer from the options given below:

- A-II, B-I, C-III, D-IV
- A-II, B-I, C-IV, D-III
- A-IV, B-III, C-I, D-II
- A-I, B-II, C-III, D-IV

122. Read the following statements w.r.t population attributes.

- A. A population at any given time is composed of individuals of different ages.
- B. The size of a population for any species is a static parameter.
- C. Immigration is the number of individuals of the population who left the habitat and gone elsewhere during a particular time period.

Choose the *most appropriate* answer from the options given below:

- (1) A and B are correct and C is incorrect.
- (2) A and C are correct and B is incorrect.
- (3) A is correct and B and C are incorrect.
- (4) B is correct and A and C are incorrect.

123. If fatty acids were to be respired they would first be degraded to _____ and enter the respiratory pathway.

- (1) PGAL
- (2) glycerol
- (3) acetyl CoA
- (4) dihydroxyacetone phosphate

124. Given below are some statements (A-D).

- A. The Green revolution succeeded in tripling the food supply.
- B. A whole plant could be regenerated from explants.
- C. Even if the plant is infected with a virus, the meristem (apical and axillary) is free of virus.
- D. Bt toxin is produced by a bacterium called *Bacillus thuringiensis*.

Choose the **correct** statements from the options given below.

- (1) A, B and C only
- (2) A and D only
- (3) B and D only
- (4) A, B, C and D

125. Match List-I with List-II.

List-I	List-II
(A) Exon	(I) Non-coding sequence
(B) Intron	(II) Coding sequence
(C) Capping	(III) Adenylate residues
(D) Tailing	(IV) Methyl guanosine triphosphate

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-II, C-III, D-IV
- (3) A-II, B-I, C-IV, D-III
- (4) A-II, B-IV, C-I, D-III

126. Read the following and identify the **correct** statements.

- A. Dinoflagellates have two flagella; one lies tangentially and the other longitudinally in a furrow between the wall plates.
- B. In slime moulds, the plasmodium differentiates and forms fruiting bodies bearing spores at their tips.
- C. The walls of diatoms are embedded with nickel and thus the walls are indestructible.
- D. *Plasmodium* (malarial parasite) is a sporozoan.

Choose the *most appropriate* answer from the options given below:

- (1) A and B
- (2) B and C
- (3) B and D
- (4) C and D

127. Identify the **incorrect** statement(s) from the following w.r.t mosses.

- A. In the life cycle of mosses, the first stage is the protonema stage, which develops directly from a gamete.
- B. In the life cycle of mosses, second stage is the leafy stage, which develops from the secondary protonema as a lateral bud.
- C. Vegetative reproduction in mosses is by fragmentation and budding in the secondary protonema.
- D. The sporophyte in mosses is more elaborate than that in liverworts.

Choose the *most appropriate* answer from the options given below:

- (1) A only
- (2) A and B
- (3) B and D
- (4) C only

128. Match List-I with List-II.

List-I	List-II
(A) Ascomycetes	(I) <i>Colletotrichum</i>
(B) Phycomycetes	(II) <i>Neurospora</i>
(C) Basidiomycetes	(III) <i>Albugo</i>
(D) Deuteromycetes	(IV) <i>Ustilago</i>

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-I, B-II, C-III, D-IV
- (3) A-II, B-I, C-IV, D-III
- (4) A-II, B-IV, C-I, D-III

129. Given below are two statements:

Statement I: AUG has dual functions. It codes for Methionine (met) and also acts as initiator codon.

Statement II: UAA, UAG, UGG are stop terminator codons.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

130. The annual net primary productivity (dry weight) of the whole biosphere is approximately:

- (1) 55 billion tons
- (2) 170 billion tons
- (3) 155 billion tons
- (4) 165 billion tons

131. Read the following statements carefully and select the **correct** ones.

- A. Inclusion bodies are not bound by any membrane system and lie free in the nucleoplasm.
- B. All eukaryotic cells are not identical.
- C. Both the cilium and flagellum emerge from centriole like structure called the basal bodies.

D. The materials to be packaged from ER fuse with the *trans* face of the golgi apparatus and move towards the maturing face.

Choose the **most appropriate** answer from the options given below:

- (1) A and C
- (2) C and D
- (3) B and C
- (4) A and D

132. The green expanded part of the leaf with veins and veinlets is:

- (1) pulvinus.
- (2) leaf base.
- (3) petiole.
- (4) leaf blade.

133. _____ is the substrate for the enzyme beta galactosidase and it regulates switching on and off of the lac operon.

- (1) Glucose
- (2) Fructose
- (3) Galactose
- (4) Lactose

134. Identify the **correct** match w.r.t C_4 pathway in plants.

- (1) Presence of PEP - Mesophyll cell
- (2) Regeneration of PEP - Bundle sheath cell
- (3) Fixation of CO_2 by Calvin cycle - Mesophyll cell
- (4) C_4 acid formation - Bundle sheath cell

135. Read the following statements w.r.t decomposition by microbes and select how many statement(s) is/are **correct**?

- A. Detritivores break down detritus into smaller particles. This process is called fragmentation.
 - B. By the process of catabolism, water-soluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts.
 - C. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called as leaching.
 - D. The humus is degraded by some microbes and release of inorganic nutrients occur by the process known as mineralisation.
- (1) One
 - (2) Three
 - (3) Two
 - (4) Four

Botany : Section-B (Q. No. 136 to 150)

136. India now has _____ biosphere reserves, _____ national parks and _____ wildlife sanctuaries, respectively.
- (1) 25, 34, 448 (2) 448, 90, 14
(3) 34, 25, 90 (4) 14, 90, 448
137. Given below are two statements:
Statement I: Xylem fibres have highly thickened walls and obliterated peripheral lumens.
Statement II: The radial conduction of water takes place by the ray parenchymatous cells.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.
138. Identify the **incorrect** statement w.r.t G_0 phase.
- (1) It is also known as quiescent stage.
 (2) Cell of this stage remain metabolically inactive.
 (3) There is no longer proliferation in the cells which are present in this stage, unless called to do so.
 (4) Heart cells in animals exhibit in this stage.
139. Identify the **incorrect** statement(s) from the following.
- A. Both the species benefit in mutualism and both lose in competition in their interactions with each other.

- B. Biological control methods are based on the ability of the predator to regulate prey population.
 C. Totally unrelated species could not compete for the same resource.
 D. The life cycles of endoparasites are more complex because of their extreme specialisation.

Choose the *most appropriate* answer from the options given below:

- (1) A and B (2) B only
(3) C only (4) C and D

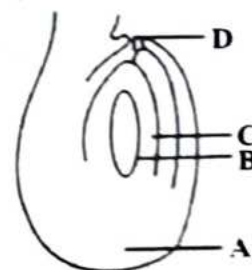
140. Match List-I with List-II.

List-I	List-II
(A) Epipetalous	(I) Pea
(B) Polyadelphous	(II) Brinjal
(C) Diadelphous	(III) Citrus
(D) Monadelphous	(IV) China rose

Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-III, D-II
 (2) A-IV, B-II, C-III, D-I
 (3) A-II, B-I, C-IV, D-III
 (4) A-II, B-III, C-I, D-IV

141. Identify the parts labelled A, B, C and D in the given figure and select the **correct** option.



A	B	C	D
(1) Chalazal pole	Female gametophyte	Embryo sac	Micropyle
(2) Chalazal pole	Embryo sac	Nucellus	Micropyle
(3) Micropyle	Egg	Embryo sac	Chalazal pole
(4) Micropyle	Nucellus	Embryo sac	Chalazal pole

142. Match List-I with List-II.

List-I	List-II
(A) Host in brood parasitism	(I) Compete for zooplanktons in the lake
(B) Barnacles growing on the back of a whale	(II) Commensalism
(C) Visiting flamingoes and resident fishes in shallow South American lakes	(III) Crow
(D) Fig and pollinator wasp	(IV) Co-evolution

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-I, B-IV, C-II, D-III
- (3) A-III, B-II, C-I, D-IV
- (4) A-I, B-II, C-III, D-IV

143. Among the vertebrates, maximum number of species is observed for _____.

- (1) fishes
- (2) reptiles
- (3) birds
- (4) mammals

144. Match List-I with List-II.

List-I	List-II
(A) Auxin	(I) Tobacco
(B) Gibberellin	(II) Rice
(C) Cytokinin	(III) Orange
(D) Ethylene	(IV) Oat

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-I, B-IV, C-II, D-III
- (3) A-IV, B-II, C-I, D-III
- (4) A-I, B-II, C-III, D-IV

145. Arrange the sequence of crosses needed to achieve a genotype and phenotype ratio of 1 : 2 : 1 in F_2 of Snapdragon flowers exhibiting incomplete dominance:

- (a) RR (Red flower)
- (b) Rr (Pink flower)
- (c) rr (White flower)

Choose the correct sequence of crosses from the options given below:

- (1) $RR \times Rr \rightarrow Rr$ and Rr
- (2) $RR \times rr \rightarrow Rr$ and Rr
- (3) $rr \times Rr \rightarrow rr$ and Rr
- (4) $Rr \times RR \rightarrow RR$ and Rr

146. Study the following statements

- (a) A homozygous tall plant (TT) is crossed with homozygous dwarf (tt), the tall progeny = 100%
- (b) A heterozygous tall selfed, tall : dwarf = 75% : 25%
- (c) A heterozygous tall crossed with homozygous dwarf, the tall : dwarf is 1 : 1.
- (d) A heterozygous tall crossed with homozygous; tall : dwarf = 100 : 0.

- (1) (a) and (b) are correct
- (2) (b) and (c) are correct
- (3) (a), (b) and (c) are correct
- (4) All are correct

147. Which of the following statements are correct?

- (a) Alien species invasions cause the loss of biodiversity.
- (b) When alien species are introduced, some of them turn invasive.
- (c) Alien species cause decline or extinction of indigenous species.
- (d) The introduction of Nile perch into Lake Victoria caused over 2000 species of cichlid fish, which are found there, to go extinct.

Select the correct option:

- (1) (a), (b) and (c)
- (2) (a), (c) and (d)
- (3) (b), (c) and (d)
- (4) (b) and (c)

148. How many statements are correct?

- (a) Tropical environments promote niche specialisation.
- (b) Tropical environments have greater species diversity.
- (c) Tropical environments, unlike temperate ones, are more seasonal.
- (d) Tropical environments, unlike temperate ones, are relatively less constant and predictable.

Select the correct option:

- (1) One
- (2) Two
- (3) Three
- (4) Four



149. Identify the wrong statement:

- (1) The probability of homozygous dwarf progeny formed in a cross $TT \times TT$ is zero.
- (2) The probability of homozygous dwarf formed in $TT \times Tt$ is 0.
- (3) The heterozygous tall progeny formed in a cross $TT \times Tt$ is 25%.
- (4) The homozygous tall progeny formed in a cross involving $TT \times TT = 1$

150. Study the given populations and choose the correct answer in relation to species diversity.

Population	Species	Group	Individuals
Population A	I	Mammals	3
	II	Birds	2
	III	Amphibians	2
Population B	I	Mammals	2
	II	Mammals	2
	III	Amphibians	1
Population C	I	Mammals	3
	II	Mammals	2
	III	Mammals	1

Maximum diversity

Minimum diversity

- | | |
|------------------|--------------|
| (1) Population B | Population C |
| (2) Population A | Population C |
| (3) Population A | Population B |
| (4) Population B | Population A |

Zoology : Section-A (Q. No. 151 to 185)

151. In which of the following applications, do we use biotechnology?

- (1) Production of insulin
- (2) Formation of vaccines
- (3) Production of pest resistant crops
- (4) All of these

152. Match List-I with List-II:

List-I	List-II
(A) P wave	(I) Heart muscles are electrically silent
(B) QRS complex	(II) Repolarisation of ventricles
(C) T wave	(III) Depolarisation of atria
(D) T-P gap	(IV) Depolarisation of ventricles

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-I, C-IV, D-II
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-IV, C-II, D-I

153. Which of the following connective tissue stores fat in the body and provides insulation?

- (1) Bone
- (2) Blood
- (3) Adipose tissue
- (4) Tendons

154. Given below are two statements:

Statement I: Spores are remains of hard parts of life forms found in rocks.

Statement II: Homology indicates common ancestry.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

155. Choose the **correct** order for the transmission of a nerve impulse from one neuron to another through synapses.

- A. Movement of synaptic vesicle towards pre-synaptic membrane.
- B. Release of neurotransmitters in the synaptic cleft.
- C. Axon terminal receives impulse.
- D. Fusion of synaptic vesicles with the plasma membrane of pre-synaptic membrane.
- E. Binding of neurotransmitters with specific receptors.

- (1) A → B → D → E → C
- (2) C → A → D → B → E
- (3) A → E → D → B → C
- (4) C → A → B → D → E

156. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Hypothalamic hormones regulate the synthesis and secretion of pituitary hormones.

Reason R: The hypothalamus produces only releasing hormones that control the release of hormones from the pituitary gland.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

157. Which of the following cells are present in the interstitial spaces (outside the seminiferous tubule)?

- (1) Male germ cells
- (2) Sertoli cells
- (3) Leydig cells
- (4) Follicular cells

158. Given below are some statements (A-D).

- A. On average, 1100-1200ml of blood is filtered by the kidneys per minute.
- B. The amount of filtrate formed by the kidneys per hour is called glomerular filtration rate (GFR).
- C. Nearly 99 per cent of the filtrate has to be reabsorbed by the renal tubules.
- D. Reabsorption of water occurs actively in the initial segments of the nephron.

Choose the **incorrect** statements from the options given below.

- (1) A and C only
- (2) A and D only
- (3) C and D only
- (4) B and D only

159. Which of the following organisms shows metagenesis?

- (1) *Ctenoplana*
- (2) *Taenia*
- (3) *Obelia*
- (4) *Ancylostoma*

160. Match the columns:

Column-I	Column-II
(a) Appearance of dry, scaly lesions on various parts of the body such as skin nails and scalp	(i) <i>Entamoeba histolytica</i>
(b) Chronic inflammation of the lymphatic vessel of lower limbs	(ii) <i>Ascaris lumbricoides</i>
(c) Fever, chills, cough, headache and in severe cases the lips and fingernails may turn grey to bluish in colour	(iii) <i>Haemophilus influenzae</i>
(d) Constipation, abdominal pain and cramps, stool with excess mucous and blood clots	(iv) <i>Wuchereria bancrofti</i>
(e) Internal bleeding, muscular pain, fever, anaemia and blockage of intestinal passage	(v) <i>Microsporium</i>

Choose the **correct** answer from the options given below:

- (1) (a)-(v), (b)-(iv), (c)-(iii), (d)-(i), (e)-(ii)
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(v), (e)-(i)
- (3) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv), (e)-(v)
- (4) (a)-(iv), (b)-(iii), (c)-(v), (d)-(ii), (e)-(i)

161. What is the partial pressure of CO_2 in alveoli and tissues, respectively?

- (1) 40 mm Hg and 45 mm Hg
- (2) 0.3 mm Hg and 40 mm Hg
- (3) 45 mm Hg and 40 mm Hg
- (4) 40 mm Hg and 95 mm Hg

162. Given below are some statements (A-E).

- A. Skeletal muscle tissue is closely attached to skeletal bones.
- B. The muscle fibres are composed of numerous fine fibrils, called myofibrils.
- C. Cell junctions hold smooth muscles together.
- D. Neuroglia make up more than one-half the volume of neural tissue in our body.
- E. The main function of compound epithelium is to provide protection.

Choose the **correct** statements from the options given below.

- (1) A, B and D only
- (2) B, C and E only
- (3) C, D and E only
- (4) A, B, C, D and E

163. Which of the following options **correctly** describes the elution process of biotechnology?

- (1) Separating the restricted DNA fragments on agarose gel.
- (2) Staining the separate DNA fragments with ethidium bromide.
- (3) Cutting out of the separated band of DNA from the agarose gel and extracting them from the gel piece.
- (4) Constructing DNA by joining the purified DNA fragments to the cloning vector.

164. Which of the following hormones is responsible for the release of the ovum?

- (1) Melanocyte-stimulating hormone (MSH)
- (2) Oxytocin
- (3) Luteinising hormone (LH)
- (4) Prolactin

165. Given below are two statements:

Statement I: Inorganic compounds like sulphate, phosphate, etc., are seen in the acid-insoluble fraction.

Statement II: Lecithin is a phospholipid.

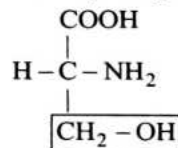
In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

166. Choose the **mismatched** option.

- (1) Evolutionary biology - Study of the history of life forms on earth.
- (2) Branching descent - Key concept of the Darwinian theory of evolution.
- (3) Saltation - Single-step large mutation.
- (4) *Homo erectus* - Arose in Africa and moved across continents and developed into distinct races.

167. Identify the given structure.



- | | |
|-------------|------------|
| (1) Glycine | (2) Ribose |
| (3) Alanine | (4) Serine |

168. Match List-I with List-II:

List-I	List-II
(A) <i>Bacillus thuringiensis</i>	(I) Cloning vector
(B) <i>Agrobacterium tumefaciens</i>	(II) Infects tobacco plant roots
(C) <i>Meloidogyne incognita</i>	(III) Resistance to insects
(D) RNAi	(IV) Gene silencing

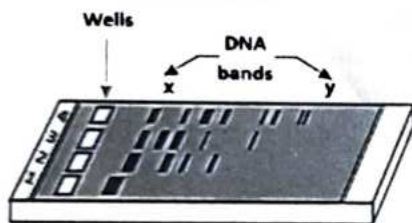
Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
- (2) A-I, B-II, C-III, D-IV
- (3) A-III, B-IV, C-II, D-I
- (4) A-IV, B-II, C-I, D-III

169. Chordae tendineae are found in:

- (1) ventricles of brain.
- (2) joints of legs.
- (3) ventricles of heart.
- (4) atria of heart.

170. Study the given figure carefully and select the **incorrect** statements.



- A. It represents a typical agarose gel electrophoresis.
 - B. The smallest DNA bands are formed at X and the largest DNA bands are formed at Y.
 - C. The separated DNA fragments can be visualised after staining in the visible light.
- (1) A and B only
 - (2) B and C only
 - (3) A and C only
 - (4) None of these

171. Given below are some statements (A-D).

- A. Parietal bone and the temporal bone of the skull are joined by a fibrous joint.
- B. The first vertebra is the axis that articulates with the occipital condyles.
- C. The 9th and 10th pairs of ribs are called the floating ribs.
- D. Glenoid cavity is a depression to which the thigh bone articulates.

Choose the **incorrect** statements from the options given below.

- (1) A and B only
- (2) A, B and C only
- (3) B, C and D only
- (4) A, B, C and D

172. Match List-I with List-II:

List-I	List-II
(A) Amphibia	(I) Air bladder
(B) Mammals	(II) Cartilaginous endoskeleton
(C) Chondrichthyes	(III) Mammary glands
(D) Osteichthyes	(IV) Dual habitat

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-IV, B-II, C-I, D-III

173. Which of the following drugs is being excessively taken by some sports persons nowadays?

- (1) Heroine
- (2) Barbiturates
- (3) Cannabinoids
- (4) Morphine

174. Spermiation is the process of the release of sperms from:

- (1) seminiferous tubules.
- (2) vas deferens.
- (3) epididymis.
- (4) prostate gland.

175. During insertional inactivation, the presence of a chromogenic substrate gives blue-coloured colonies if the plasmid in the bacteria does not have an insert. The blue colour is produced by the enzyme:

- (1) α -glucosidase.
- (2) restriction endonuclease.
- (3) β -galactosidase.
- (4) *Taq* polymerase.

176. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Enzymes generally function in a narrow range of temperature and pH.

Reason R: Enzymes show the highest activity at a particular temperature and pH which is the same for all groups of enzymes.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

177. Given below are two statements:

Statement I: Nearly 45 to 50 million MTPs are performed in a year all over the world.

Statement II: MTPs are considered relatively safe during the 3rd trimester.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

178. Select the mismatched pair.

- (1) Inulin – Fructose
- (2) Cellulose – Heteropolymer
- (3) Starch – Helical secondary structure
- (4) Chitin – Complex polysaccharide

179. Match List-I with List-II:

List-I		List-II	
(A)	Camouflage	(I)	Winter sleep
(B)	Aestivation	(II)	Ability to change colour to hide from their enemies
(C)	Hibernation	(III)	Summer sleep
(D)	Human liver fluke	(IV)	Two intermediate hosts

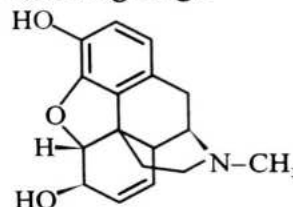
Choose the correct answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
- (2) A-III, B-II, C-I, D-IV
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-I, C-III, D-IV

180. Identify the technique based on antigen-antibody interaction.

- (1) Polymerase chain reaction
- (2) ELISA
- (3) Transformation
- (4) Electrophoresis

181. The given diagram represents which of the following drugs?



- (1) Morphine
- (2) Cannabinoids
- (3) Cocaine
- (4) Benzodiazepine

182. Match List-I with List-II:

List-I	List-II
(A) Cleavage	(I) Fertilisation
(B) Morula	(II) Mitotic divisions
(C) Ampullary region	(III) Endometrium
(D) Implantation	(IV) 8-16 celled stage

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-III, B-I, C-II, D-IV
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-IV, C-I, D-III

183. Given below are some statements (A-E).

- A. In Urochordata, the notochord is persistent throughout their life.
- B. In Chondrichthyes, the mouth is located ventrally.
- C. The Osteichthyes are both marine and freshwater fishes.
- D. Osteichthyes have placoid scales.
- E. *Salpa* is an example of cephalochordates.

Choose the **incorrect** statements from the options given below.

- (1) A, B and C only
- (2) A, D and E only
- (3) B, C and D only
- (4) A, C and E only

184. Lysine is an essential amino acid because;

- (1) it is very rare.
- (2) it has high nutritive value.
- (3) it is an important constituent of all proteins.
- (4) it is not formed in the body and has to be provided through diet.

185. Match List-I with List-II:

List-I	List-II
(A) <i>Ramapithecus</i>	(I) 75,000-10,000 years ago
(B) <i>Homo erectus</i>	(II) 1,00,000-40,000 years back
(C) Neanderthal man	(III) 15mya
(D) <i>Homo sapiens</i>	(IV) 1.5mya

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-III, B-IV, C-II, D-I
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-IV, C-I, D-III

Zoology : Section-B (Q. No. 186 to 200)

186. Given below are two statements:

Statement I: Each ovary is connected to the pelvic wall and uterus by ligaments.

Statement II: The presence or absence of a hymen is a reliable indicator of virginity.

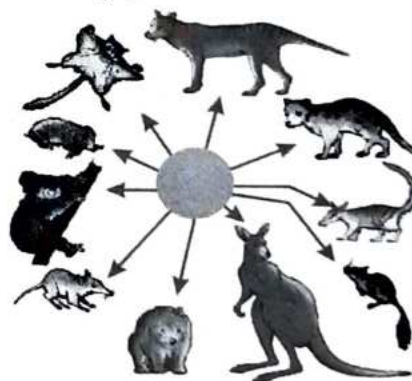
In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

187. Choanocytes are present in:

- (1) *Physalia*.
- (2) *Sycon*.
- (3) *Pleurobrachia*.
- (4) *Echinus*.

188. Study the figure given below and answer the following question.



Which of the following animals are depicted in the figure?

- (1) Placental mammals
- (2) Darwin's finches
- (3) Australian marsupials
- (4) Egg-laying mammals

189. Which of the following statements is **true** about RBCs in humans?

- (1) They carry about 20-25 per cent of CO_2 .
- (2) They transport 99.5 percent of O_2 .
- (3) They transport about 80 per cent oxygen and the remaining 20 per cent of it is transported in a dissolved state in blood plasma.
- (4) They do not carry CO_2 at all.

190. Given below are some statements (A-D).

- A. Each haemoglobin molecule can carry a maximum of four molecules of O_2 .
- B. The binding of oxygen with haemoglobin is primarily related to the partial pressure of O_2 .
- C. O_2 can reversibly bind with haemoglobin to form oxyhaemoglobin.
- D. Every 100 ml of deoxygenated blood delivers approximately 20 ml of CO_2 to the alveoli.

Choose the **correct** statements from the options given below.

- (1) A, B and C only
- (2) A and D only
- (3) B and D only
- (4) C and D only

191. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: During muscle contraction, actin filament slides over myosin filament.

Reason R: A band remains unchanged in muscle contraction.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

192. Match List-I with List-II:

List-I	List-II
(A) Condoms	(I) Natural method of contraception
(B) Lactational amenorrhea	(II) Barrier method of contraception
(C) Multiload 375	(III) Non medicated IUD
(D) Lippes loop	(IV) Copper releasing IUD

Choose the **correct** answer from the options given below:

- (1) A- IV, B-III, C-II, D-I
- (2) A-IV, B-I, C-II, D-III
- (3) A-II, B-III, C-I, D-IV
- (4) A-II, B-I, C-IV, D-III

193. Which of the following statements is **true** for proteins?

- (1) Homopolymer containing strings of fatty acids.
- (2) Heteropolymer containing strings of fatty acids.
- (3) Homopolymer containing strings of amino acids.
- (4) Heteropolymer containing strings of amino acids.

194. In *EcoR* I, 'R' refers to the:

- (1) name of the genus.
- (2) name of the species.
- (3) name of the strain.
- (4) order in which the enzymes were isolated from the strain of bacteria.

195. Given below are some statements (A-E).

- A. Hormones are non-nutrient chemicals.
- B. The posterior pituitary is under the direct neural regulation of the hypothalamus.
- C. Melanocytes regulate the pigmentation of the skin.

- D. Oxytocin acts on the smooth muscles of our body and stimulates their contraction.
E. Thyrocalcitonin is a steroid hormone.
Choose the **correct** statements from the options given below.

- (1) A, B, C and D only
- (2) A, D and E only
- (3) B, C and D only
- (4) D and E only

196. Typhoid fever in human beings is caused by:

- (1) *Plasmodium vivax*.
- (2) *Trichophyton*.
- (3) *Salmonella typhi*.
- (4) Rhinovirus.

197. Which of the following is the **correct** pathway for passage of urine in humans?

- (1) Kidney → Ureter → Bladder → Urethra
- (2) Renal vein → Ureter → Bladder → Urethra
- (3) Bladder → Ureter → Kidney → Urethra
- (4) Urethra → Kidney → Bladder → Ureter

198. Which of the following are the regions of the body of the cockroach?

- (1) Head, thorax and abdomen
- (2) Head and thorax only
- (3) Head and abdomen only
- (4) Head, body and tail

199. In human beings, the membrane of the erythrocyte has approximately;

- (1) 40 percent lipids and 52 percent carbohydrates.
- (2) 40 percent protein and 52 percent lipids.
- (3) 40 percent lipids and 52 percent proteins.
- (4) 40 percent protein and 52 percent carbohydrates.

200. Match List-I with List-II:

List-I	List-II
(A) Between the humerus and pectoral girdle	(I) Gliding joint
(B) Between the carpals	(II) Saddle joint
(C) Between carpal and metacarpal of thumb	(III) Pivot joint
(D) Between atlas and axis	(IV) Ball and socket joint

Choose the **correct** answer from the options given below:

- (1) A- IV, B-III, C-II, D-I
- (2) A-IV, B-I, C-II, D-III
- (3) A-II, B-III, C-I, D-IV
- (4) A-II, B-I, C-IV, D-III